

Stoichiometry of a saturated fluorite nanoparticle

Exact lattice counts for axis-aligned nanocubes and boxes

Abstract

We consider a fluorite (AB_2) nanoparticle in the idealised limit where every cation A , including those at the surface, retains its full eightfold coordination by anions B . In this model the anions form a simple-cubic lattice and the cations occupy the body centres of one checkerboard colour of the anion cubes. The stoichiometry is therefore an exact finite lattice-counting problem. We prove a bond-counting identity showing that every finite saturated particle is B -rich, show by example that the ratio is not a function of N_A alone, and derive closed forms for every axis-aligned $a \times b \times c$ box. The associated $n \times n \times n$ nanocube sequences are elementary parity-modulated polynomials; the full B -count and B -excess sequences currently have no direct OEIS match and are natural, though modest, catalogue candidates.

1 The fluorite structure, recast

In the conventional cubic cell of fluorite CaF_2 (side a), the cation A occupies the fcc positions and the anion B occupies all eight tetrahedral holes. The useful simplification is to read the structure from the anion side.

Proposition 1.1. *In fluorite, B forms a simple-cubic lattice of spacing $a/2$,*

$$B = \frac{a}{2}\mathbb{Z}^3 + \frac{a}{4}(1, 1, 1),$$

and A sits at the body centre of every second B -cube. In B -lattice units, indexing unit B -cubes by $(i, j, k) \in \mathbb{Z}^3$,

$$A = \{(i + \frac{1}{2}, j + \frac{1}{2}, k + \frac{1}{2}) : i + j + k \text{ even}\}.$$

Consequently $\text{CN}(A) = 8$ and $\text{CN}(B) = 4$ in the infinite bulk. The number densities in these units are $\rho_A = 1/2$ and $\rho_B = 1$.

Proof. The eight tetrahedral holes are $(a/4)(\pm 1, \pm 1, \pm 1)$; together with translations by $a\mathbb{Z}^3$ they form $\frac{a}{2}\mathbb{Z}^3 + (a/4)(1, 1, 1)$. Shifting the origin by $(a/4)(1, 1, 1)$ places the anions at the integer lattice. The fcc cation positions are then the body centres of exactly one checkerboard colour of the unit cubes. The coordination numbers are immediate from the cube-corner incidence. $\square$

2 The saturation model

Definition 2.1 (saturated particle). A particle is specified by a finite set S of A -sites. Its B -set is

$$B(S) = \bigcup_{A \in S} \{\text{the eight corner-}B \text{ sites of the } A \text{ cube}\}.$$

Thus every selected cation, including a surface cation, keeps its full eightfold B shell. Shared corner anions are counted once.

Let $N_A = |S|$, $N_B = |B(S)|$, and let $d_B \in \{1, 2, 3, 4\}$ be the number of selected A -cubes incident on a given included B site.

Theorem 2.2 (bond-counting identity). *For every finite saturated particle,*

$$N_B - 2N_A = \frac{1}{4} \sum_{B \in B(S)} (4 - d_B) \geq 0.$$

Equality is impossible for a nonempty finite particle, so every finite saturated particle is strictly B -rich.

Proof. Counting A - B incidences from the A side gives $8N_A$. Counting the same incidences from the B side gives $\sum_B d_B$. Hence

$$2N_A = \frac{1}{4} \sum_B d_B, \quad N_B - 2N_A = \frac{1}{4} \sum_B (4 - d_B).$$

Each summand is non-negative because a B site has at most four adjacent checkerboard A -cubes. A finite particle must have boundary B sites with $d_B < 4$. $\square$

The excess $N_B - 2N_A$ is therefore a direct measure of missing bulk neighbours at the boundary.

3 The ratio is not a function of the cation count alone

Proposition 3.1 (form-factor dependence). *Two saturated particles with the same N_A can have different N_B .*

Example 3.2. The $2 \times 2 \times 2$ axis-aligned cube has $N_A = 4$ and $N_B = 23$. The $4 \times 2 \times 1$ slab also has $N_A = 4$, but it has $N_B = 26$. Thus the ratio is 1:5.75 in one case and 1:6.50 in the other.

Consequently one should not expect a universal function $N_B = f(N_A)$. A well-defined sequence appears only after choosing a shape family and a placement convention.

4 Axis-aligned boxes

The cleanest exact family is the axis-aligned box $[0, a) \times [0, b) \times [0, c)$ of B -cubes, with the selected A -sites those for which $i + j + k$ is even.

Theorem 4.1 (box formula). *For positive integers a, b, c ,*

$$N_A = \text{ceil}\left(\frac{abc}{2}\right), \quad N_B = (a+1)(b+1)(c+1) - U,$$

where $U \in \{0, 1, 2, 3, 4\}$ is the number of enclosing-box vertices not incident to any selected A -cube. If $\varepsilon = 1$ when a, b, c are all odd and $\varepsilon = 0$ otherwise, then

$$N_B - 2N_A = (ab + ac + bc) + (a + b + c) + (1 - U - \varepsilon). \quad (1)$$

Proof. The number of even-parity triples in the $a \times b \times c$ index box is $\text{ceil}(abc/2)$, giving N_A . If all cube colours were present, the touched B -vertices would be the full grid of size $(a+1)(b+1)(c+1)$. With only one checkerboard colour, the only possible omissions are enclosing-box vertices whose unique incident in-box cube has odd parity; their count is U . Therefore $N_B = (a+1)(b+1)(c+1) - U$. Subtracting $2N_A = abc + \varepsilon$ and expanding gives Eq. (1). $\square$

The three terms in Eq. (1) have a useful interpretation for this axis-aligned family. The leading term $ab + ac + bc$ is the sum of the three coordinate projected areas of the box. The next term $a + b + c$ is an edge-order correction. The final term is a bounded parity/corner correction.

For any fixed axis-aligned aspect ratio (α, β, γ) scaled by a large parameter m , the excess is of order m^2 while N_A is of order m^3 . Thus

$$\frac{N_B}{N_A} = 2 + c_{\alpha, \beta, \gamma} N_A^{-1/3} + o(N_A^{-1/3}), \quad c_{\alpha, \beta, \gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{(\alpha\beta\gamma/2)^{2/3}}. \quad (2)$$

For the cube this gives $c = 3 \cdot 2^{2/3} \approx 4.762$.

Remark (scope). Equation (1) is an exact axis-aligned box result, not a universal surface theorem. A tempting continuum argument would inflate a region by the full cube $[-1/2, 1/2]^3$, leading to a projected-area formula for arbitrary orientations. That argument ignores that the A sites occupy only one checkerboard colour. Oblique and curved boundaries sample the checkerboard adjacency differently, so no sphere, ellipsoid, or Wulff-shape claim is made here.

5 Cubic-shell sequences and the OEIS

For the $n \times n \times n$ saturated cube, using the checkerboard phase that contains the origin cube, Theorem 4.1 gives

$$N_A(n) = \text{ceil}\left(\frac{n^3}{2}\right), \quad N_B(n) = (n+1)^3 - 2(1 + (-1)^n),$$

and

$$N_B(n) - 2N_A(n) = 3n(n+1) - \frac{3}{2}(1 + (-1)^n) = \begin{cases} 3n(n+1), & n \text{ odd}, \\ 3(n^2 + n - 1), & n \text{ even}. \end{cases}$$

The phase convention matters for odd n : the opposite checkerboard colour in the same enclosing cube has one fewer A site and a different B count. The choice above matches the $\text{ceil}(n^3/2)$ convention of A036486.

Table 1: Cubic-shell sequences and OEIS status. Direct OEIS searches on 2026-06-25 returned no match for the candidate rows.

sequence	first terms	OEIS
N_A (full)	1, 4, 14, 32, 63, 108, 172, 256, ...	A036486 $\text{ceil}(n^3/2)$
N_A odd shells	1, 14, 63, 172, 365, 666, 1099, ...	A050492 thickened cubes
N_B odd shells	8, 64, 216, 512, 1000, 1728, ...	A016743 even cubes
excess odd shells	6, 36, 90, 168, 270, 396, ...	A152746 $6 \times$ hexagonal numbers
N_B (full)	8, 23, 64, 121, 216, 339, 512, 725, ...	candidate: no direct match
excess (full)	6, 15, 36, 57, 90, 123, 168, 213, ...	candidate: no direct match
N_B even shells	23, 121, 339, 725, 1327, 2193, ...	redundant subsequence
excess even shells	15, 57, 123, 213, 327, ...	redundant subsequence

The two full candidate sequences are elementary: one is a parity-modulated cubic, the other a parity-modulated quadratic. Their possible OEIS value comes from the fluorite counting interpretation and the link to A036486, not from intrinsic number-theoretic depth. The even-shell subsequences should not be submitted separately unless an editor specifically asks for them.

6 Physical scope

The model is deliberately combinatorial. It counts the anions required to keep all selected cations eightfold coordinated in the ideal fluorite adjacency graph. It does not include electrostatics, charge neutrality mechanisms, surface relaxation, reconstruction, defects, hydration, or real surface energies. In particular, the exact box formula should not be read as a prediction that real fluorite nanoparticles prefer cube facets. Real CaF_2 morphology is governed by chemistry outside this counting model.

7 Conclusion

The robust result is exact and finite: saturated fluorite boxes have the closed forms of Theorem 4.1, and the nanocube produces simple parity-split formulas for N_A , N_B , and $N_B - 2N_A$. The result is not a universal formula depending only on N_A , nor is it a continuum Wulff theorem. As an OEIS contribution, the strongest candidate is the B -excess sequence, with the B -count included as an optional companion or comment.

A Verification

All exact formulas are checked by brute-force enumeration in `scripts/`:

- `fluorite.py`: core lattice-counting library.
- `verify_sequences.py`: cubic-shell formulas, box formulas, bond identity, non-universality example, and OEIS data preparation.
- `verify_box_formulas.py`: exact box decomposition, axis-aligned asymptotics, and a small oblique-boundary scope check.
- `oeis_check.py`: live OEIS lookup for the sequence table.
- `plot_ratio.py`: figures for exact axis-aligned box families.
- `run_all.sh`: runs the checks and writes `data/sequences.csv`.

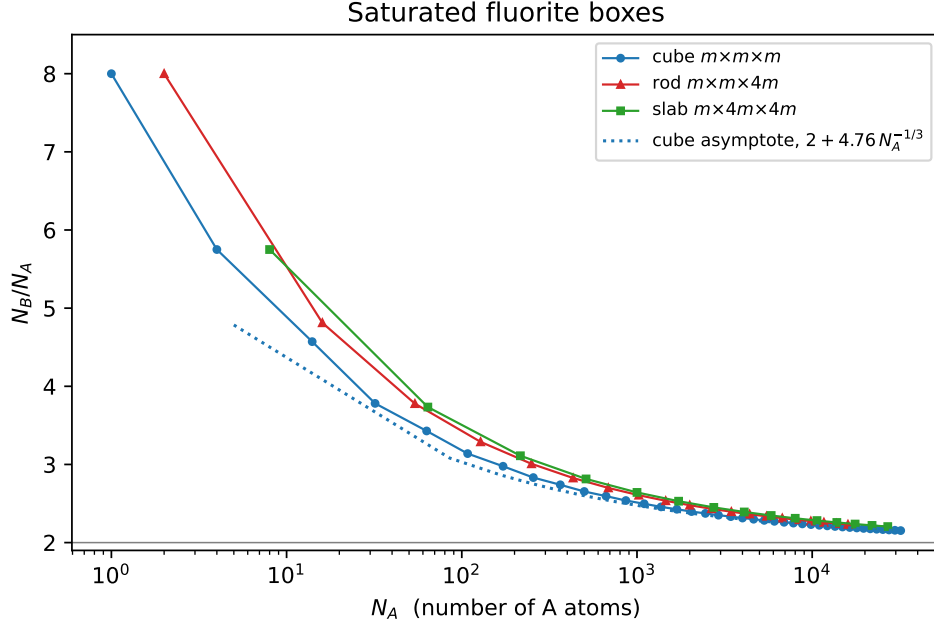


Figure 1: N_B/N_A versus N_A for exact axis-aligned box families. The ratio approaches the bulk value 2 from above, with a shape-dependent boundary coefficient.

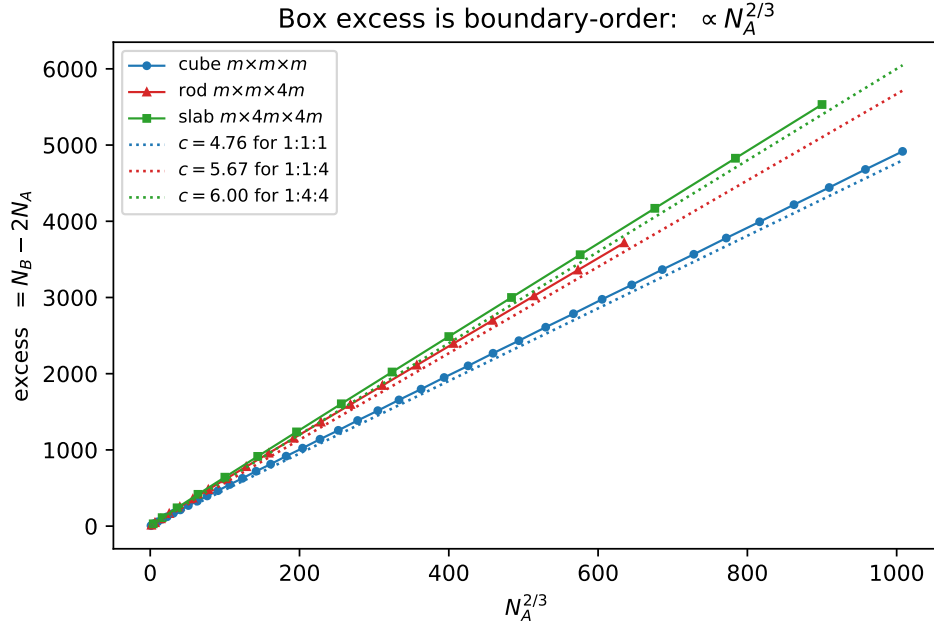


Figure 2: For fixed axis-aligned aspect ratios, the excess $N_B - 2N_A$ is a boundary-order term proportional to $N_A^{2/3}$ to leading order.